

Maharashtra Public Service Commission Assessment

Subject Name :	Assistant Professor Microbiology Group A Higher and Technical Education Department
Duration :	60
Number of Questions :	100
Total Marks :	200
Display Marks:	Yes
Maximum Instruction Time :	0
Is Section Default? :	No

Question Number : 1 Question Id : 4410092126189 Question Type : MCQ

Which of the following describes the main purpose of simple staining in microbiology?

Options :

1. To enhance the visibility of bacterial cells under a microscope
2. To identify the presence of endospores in bacteria
3. To detect bacterial motility in a wet mount
4. To differentiate between Gram-positive and Gram-negative bacteria

Question Number : 2 Question Id : 4410092130254 Question Type : MCQ

Which taxonomic rank groups similar genera together in the classification of microorganisms?

Options :

1. Class
2. Family
3. Species
4. Order

Question Number : 3 Question Id : 4410092126191 Question Type : MCQ

Which step of the Gram staining procedure differentiates Gram-positive bacteria from Gram-negative bacteria?

Options :

1. Addition of iodine
2. Counterstaining with safranin
3. Alcohol decolorization
4. Application of crystal violet

Question Number : 4 Question Id : 4410092126204 Question Type : MCQ

Which statement best describes the process of binary fission in bacterial cells during microbial growth?

Options :

1. Forms spores for reproduction
2. Reproduces by conjugation
3. Divides into two identical daughter cells
4. Forms a bud that grows into a new cell

Question Number : 5 Question Id : 4410092130247 Question Type : MCQ

What is the correct order of taxonomic ranks starting from broadest to most specific?

Options :

1. Phylum, class, domain, family, order, genus, species
2. Genus, family, order, class, species, phylum, domain
3. Kingdom, genus, order, class, family, species
4. Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species

Question Number : 6 Question Id : 4410092126185 Question Type : MCQ

Which advanced light microscopy technique is suitable for observing live bacterial cells and their internal structures without staining?

Options :

1. Phase-contrast microscopy
2. Electron microscopy
3. Bright-field microscopy
4. Compound microscopy

Question Number : 7 Question Id : 4410092126187 Question Type : MCQ

A researcher wants to study the internal ultrastructure of bacterial cells at high resolution. Which electron microscopy technique should be used, and why would this choice be more appropriate than the alternative?

Options :

1. SEM- provide surface details rather than internal structures.
2. Light microscopy – Provides high resolution internal details
3. TEM - provides internal detail due to thin sections of cells.
4. Phase contrast microscopy-Reveals ultrastructure at very high resolution

Question Number : 8 Question Id : 4410092126192 Question Type : MCQ

During Ziehl–Neelsen staining, some rod-shaped bacteria lose the carbol fuchsin after acid-alcohol treatment but appear Gram-positive. Based on this result, which of the following is the most likely interpretation?

Options :

1. Not acid-fast bacteria and likely different Gram-positive bacteria.
2. Result inconclusive with Ziehl-Neelsen staining.
3. Gram-negative bacteria misidentified by the stain.
4. Acid-fast bacteria, likely *Mycobacterium tuberculosis*.

Question Number : 9 Question Id : 4410092126197 Question Type : MCQ

Which strategy best controls a multidrug-resistant bacterial outbreak while minimizing further resistance?

Options :

1. Relying solely on high-dose antibiotics for rapid control
2. Using broad-spectrum antibiotics for all patients
3. Random antibiotic rotation
4. Targeted antibiotic therapy based on susceptibility testing

Question Number : 10 Question Id : 4410092130261 Question Type : MCQ

A hospital investigates an outbreak caused by an unknown microbe. The organism is identified using serological tests, genetic sequencing, and biochemical assays. Which taxonomic approach is being used?

Options :

1. Classical taxonomy using observable traits only
2. Single biochemical test for identification
3. Molecular taxonomy based solely on genetic analysis
4. Polyphasic taxonomy involving multiple lines of evidence

Question Number : 11 Question Id : 4410092126223 Question Type : MCQ

Which process occurs in microorganisms when oxygen is absent and results in the production of energy and organic end products such as ethanol or lactic acid?

Options :

1. Fermentation
2. Phototrophy
3. Nitrogen fixation
4. Respiration

Question Number : 12 Question Id : 4410092126228 Question Type : MCQ

Which factor does NOT directly affect the rate of an enzyme-catalyzed reaction?

Options :

1. Type of substrate
2. Enzyme concentration
3. Temperature
4. pH

Question Number : 13 Question Id : 4410092126213 Question Type : MCQ

Which factor most significantly enhances the thermal stability of microbial DNA compared to RNA?

Options :

1. Presence of deoxyribose sugar and double-stranded helix
2. Presence of ribose sugar and single-stranded structure
3. Use of uracil instead of thymine in the nucleic acid
4. Increased number of phosphodiester bonds

Question Number : 14 Question Id : 4410092126215 Question Type : MCQ

Which statement best describes anabolic pathways in microbial metabolism?

Options :

1. They synthesize complex molecules using energy.
2. They release heat by decomposing cellular components.
3. They directly transfer electrons during respiration.
4. They generate energy by breaking down complex molecules.

Question Number : 15 Question Id : 4410092126217 Question Type : MCQ

Which process best describes ATP synthesis during aerobic respiration in microbial cells?

Options :

1. ATP is produced via oxidative phosphorylation
2. ATP is produced through fermentation only
3. ATP is produced via glycolytic phosphorylation only
4. ATP is produced by cytoplasmic photophosphorylation

Question Number : 16 Question Id : 4410092126211 Question Type : MCQ

A microbiologist is designing a recombinant protein for expression in a thermophilic microorganism. Which strategy is likely to enhance the protein's thermal stability while retaining its biological function?

Options :

1. Add random coil regions to increase flexibility
2. Increase hydrophobic residues in the protein core
3. Substitute all surface-exposed serine with alanine
4. Remove ionic bonds

Question Number : 17 Question Id : 4410092126218 Question Type : MCQ

A microbiologist wants to enhance the production of polyhydroxyalkanoates (PHAs) in a bacterial culture. Which metabolic strategy should be prioritized to maximize PHA accumulation during lipid metabolism?

Options :

1. Limiting nitrogen while providing excess carbon
2. Limiting both nitrogen and carbon
3. Providing balanced nitrogen and carbon
4. Supplying excess nitrogen and carbon together

Question Number : 18 Question Id : 4410092126229 Question Type : MCQ

A researcher observes that the activity of a key metabolic enzyme in a bacterium rapidly decreases when intracellular ATP concentration rises, even though the enzyme's gene expression and protein levels remain unchanged. Which regulatory mechanism most likely explains this observation?

Options :

1. Proteolytic cleavage of the enzyme into inactive fragments
2. Transcriptional repression of the enzyme gene
3. Covalent modification through enzyme phosphorylation
4. Allosteric inhibition by ATP molecules binding to the enzyme

Question Number : 19 Question Id : 4410092126231 Question Type : MCQ

A pharmaceutical company is seeking to discover a novel antibiotic with minimal cytotoxicity to human cells. Which strategy most effectively utilizes microbial secondary metabolites in this context?

Options :

1. Screening endophytic actinomycetes for new bioactive compounds
2. Using only primary metabolites from common soil bacteria
3. Developing compounds that disrupt eukaryotic ribosomes
4. Focusing exclusively on optimizing fermentation conditions

Question Number : 20 Question Id : 4410092126232 Question Type : MCQ

How can X-ray crystallography be optimized for a radiation-sensitive enzyme available in very small amounts?

Options :

1. Use neutron diffraction
2. Increase crystal thickness and collect data at room temperature
3. Use cryo-cooling and synchrotron radiation
4. Use standard X-ray sources without cryo-cooling

Question Number : 21 Question Id : 4410092126239 Question Type : MCQ

Which process describes the formation of organic molecules from simple inorganic substances on early Earth?

Options :

1. Biotic synthesis
2. Chemosynthesis
3. Photosynthesis
4. Abiotic synthesis

Question Number : 22 Question Id : 4410092126242 Question Type : MCQ

Which key finding did the Miller experiment demonstrate about the origin of cells?

Options :

1. Cells can spontaneously form from amino acid solutions.
2. Complex proteins are produced directly from atmospheric gases.
3. Amino acids can form from simple gases under early Earth conditions.
4. DNA was present in the original gas mixture.

Question Number : 23 Question Id : 4410092126240 Question Type : MCQ

Which statement best describes the relationship between organic monomers and the evolution of the first unicellular organisms?

Options :

1. Unicellular life evolved before monomers
2. Monomers formed polymers that led to cells
3. Polymers formed before monomers
4. Monomers formed inorganic structures

Question Number : 24 Question Id : 4410092126252 Question Type : MCQ

Which of the following processes occurs during RNA processing in eukaryotic cells?

Options :

1. Removal of exons from pre-mRNA
2. Translation of mRNA into protein
3. DNA methylation of the RNA strand
4. Addition of a 5' cap to the pre-mRNA

Question Number : 25 Question Id : 4410092155263 Question Type : MCQ

A patient's cancer cells show frequent chromosomal translocations. Which genetic process is most likely defective?

Options :

1. DNA repair pathways
2. RNA processing
3. Ribosome assembly
4. Protein synthesis

Question Number : 26 Question Id : 4410092126235 Question Type : MCQ

A microbial geneticist is studying a bacterial population exhibiting an unexpected phenotypic ratio in a dihybrid cross, deviating from the classic 9:3:3:1 Mendelian ratio. Which scenario best explains this deviation?

Options :

1. Acquisition of traits through horizontal gene transfer
2. Incomplete dominance at a single gene locus
3. Random fertilization of gametes
4. Epistasis between two genes affecting metabolic pathways

Question Number : 27 Question Id : 4410092126236 Question Type : MCQ

Two bacterial populations are exposed to changing antibiotic environments. Population A has high genetic variation, while Population B has low variation. Which outcome is most likely over generations?

Options :

1. Population A will adapt more rapidly to antibiotics than Population B
2. Neither population will adapt to antibiotics over generations
3. Population B will adapt more rapidly to antibiotics than Population A
4. Both populations will adapt at the same rate to antibiotics

Question Number : 28 Question Id : 4410092126245 Question Type : MCQ

Which molecular feature most likely explains the unusually high DNA melting temperature of a thermophilic bacterium?

Options :

1. High Adenine-Thymine content
2. High guanine-cytosine content
3. Substitution of uracil for thymine
4. Absence of hydrogen bonds

Question Number : 29 Question Id : 4410092126246 Question Type : MCQ

Which outcome is most likely to occur if the 3'→5' exonuclease activity of DNA polymerase III is disrupted?

Options :

1. Reduced replication speed due to slower enzyme activity
2. Increased rate of replication errors due to loss of proofreading
3. Enhanced lagging strand synthesis due to altered Okazaki fragment
4. Impaired initiation of replication at the origin

Question Number : 30 Question Id : 4410092155264 Question Type : MCQ

In immunogenetics, which region of an antibody gene is responsible for generating diversity in antigen binding?

Options :

1. Leader sequence
2. Constant region
3. Signal peptide
4. Variable region

Question Number : 31 Question Id : 4410092126100 Question Type : MCQ

The primary function of the immune system in immunological surveillance is to:

Options :

1. Detect and eliminate abnormal or malignant cells
2. Maintain normal blood glucose levels
3. Regulate body temperature during infection
4. Transport oxygen to peripheral tissues

Question Number : 32 Question Id : 4410092126102 Question Type : MCQ

Which of the following best describes a heterophile antigen?

Options :

1. An antigen that is found in different species but induces similar immune responses
2. An antigen that is unique to one species and not found in any other species
3. An antigen found only in bacteria and not found in other micro organisms
4. An antigen that never induces an immune response in any host but can induce secondary immune response

Question Number : 33 Question Id : 4410092126110 Question Type : MCQ

A bacterial isolate forms mucoid colonies and resists neutrophil uptake unless opsonized. Which surface structure explains this virulence property?

Options :

1. Polysaccharide capsule
2. Flagellar filament
3. Type IV pilus
4. Lipid A endotoxin

Question Number : 34 Question Id : 4410092126114 Question Type : MCQ

In a sandwich ELISA, the antigen is captured between a capture antibody and a detection antibody. The principal advantage of using two antibodies is to improve assay:

Options :

1. Specificity by requiring dual antibody recognition
2. Sensitivity by reducing background substrate noise
3. Speed by skipping the blocking incubation step
4. Cost by eliminating need for enzyme conjugate

Question Number : 35 Question Id : 4410092126099 Question Type : MCQ

Follicular dendritic cells differ from conventional dendritic cells mainly in that they:

Options :

1. Trap native antigen-antibody complexes via Fc receptors
2. Process antigen and migrate to lymph node paracortex
3. Express MHC class II for naive T cell priming
4. Originate from bone marrow myeloid progenitor cells

Question Number : 36 Question Id : 4410092126101 Question Type : MCQ

A hapten becomes immunogenic only after conjugation to a carrier protein because it lacks:

Options :

1. Sufficient molecular size for peptide presentation
2. Epitope diversity for B cell receptor binding
3. Chemical reactivity for covalent bond formation
4. Foreignness required to trigger innate response

Question Number : 37 Question Id : 4410092126105 Question Type : MCQ

Original antigenic sin (immune imprinting) refers to the tendency of the immune system, upon exposure to an antigenically related viral strain, to:

Options :

1. Preferentially boost antibodies against the first strain encountered
2. Generate entirely new antibodies against the current strain
3. Suppress memory responses from prior viral exposures
4. Activate innate immunity instead of adaptive immunity

Question Number : 38 Question Id : 4410092126121 Question Type : MCQ

FISH probes used for detecting bacterial pathogens directly in clinical specimens commonly target:

Options :

1. 16S ribosomal RNA sequences
2. Bacterial cell wall peptidoglycan
3. Plasmid-encoded resistance genes
4. Outer membrane lipopolysaccharide

Question Number : 39 Question Id : 4410092126122 Question Type : MCQ

Which bacterial infection is classically characterized by the acute formation of a tough, grayish pseudomembrane tightly adhering to the tonsils and pharynx?

Options :

1. Streptococcal pharyngitis
2. Diphtheria
3. Pertussis
4. Epiglottitis

Question Number : 40 Question Id : 4410092126124 Question Type : MCQ

Antigenic shift in influenza A virus, responsible for pandemics, occurs mainly through:

Options :

1. Reassortment of genome segments between two strains
2. Point mutations accumulating in surface glycoproteins
3. Recombination within a single continuous RNA genome
4. Loss of neuraminidase gene during viral replication

Question Number : 41 Question Id : 4410092126127 Question Type : MCQ

Routine stool microscopy for suspected *Entamoeba histolytica* infection primarily demonstrates:

Options :

1. Cysts with four nuclei or trophozoites
2. Eggs with characteristic operculum structure
3. Larvae showing rhabditiform morphology
4. Proglottids containing gravid uterine branches

Question Number : 42 Question Id : 4410092126139 Question Type : MCQ

Which of the following best describes the main purpose of administering immune sera in medical microbiology?

Options :

1. To destroy host immune cells and suppress infection
2. To induce immunological memory against future infections
3. To provide immediate passive immunity against specific pathogens
4. To stimulate the production of long-lasting antibodies

Question Number : 43 Question Id : 4410092126123 Question Type : MCQ

Diphtheria toxin and *Pseudomonas aeruginosa* exotoxin A both inhibit protein synthesis via ADP-ribosylation of elongation factor-2, but the two toxins primarily differ in their:

Options :

1. NAD-dependent donor molecule used in reaction
2. Target substrate for the ADP-ribosylation reaction
3. Catalytic subunit mechanism of enzymatic action
4. Receptor-binding domain determining host cell tropism

Question Number : 44 Question Id : 4410092126125 Question Type : MCQ

Which of the following viruses is the most common cause of congenital infection worldwide?

Options :

1. Enterovirus
2. Herpes simplex virus type 1
3. Cytomegalovirus
4. Varicella-zoster virus

Question Number : 45 Question Id : 4410092126133 Question Type : MCQ

Waksman's systematic soil-screening program leading to streptomycin discovery was based on:

Options :

1. Screening soil actinomycetes for antibiosis against pathogens
2. Screening soil bacteria for resistance to known antibiotics
3. Screening fungal isolates for toxin production ability
4. Screening clinical isolates for biofilm forming capacity

Question Number : 46 Question Id : 4410092126138 Question Type : MCQ

Formaldehyde treatment converts bacterial exotoxin to toxoid mainly by:

Options :

1. Cross-linking free amino groups, abolishing toxicity but retaining epitopes
2. Denaturing protein tertiary structure and retain primary structure
3. Cleaving the toxin into inactive small peptide fragments having approximately 10-15 amino acids
4. Selective removal of the receptor-binding domain while leaving the catalytic domain intact.

Question Number : 47 Question Id : 4410092126151 Question Type : MCQ

A soil bacterium utilizes amino acids released by a neighboring microbe without affecting the growth of the producer. This interaction is an example of:

Options :

1. Mutualism
2. Commensalism
3. Parasitism
4. Competition

Question Number : 48 Question Id : 4410092126159 Question Type : MCQ

What primary mechanism drives rapid genome plasticity and the sudden acquisition of stress tolerance in soil bacteria?

Options :

1. Horizontal gene flow
2. Spontaneous mutation
3. Chromosome inversion
4. Ribosomal frameshift

Question Number : 49 Question Id : 4410092126146 Question Type : MCQ

In soil ecosystems, the rhizosphere shows higher microbial density than bulk soil mainly due to:

Options :

1. Higher oxygen availability near roots
2. Release of root exudates as nutrient source
3. Lower soil pH near roots
4. Greater soil porosity near roots

Question Number : 50 Question Id : 4410092126149 Question Type : MCQ

In soil microbial ecology, Which abiotic factor most directly limits microbial activity during dry periods by restricting water availability to cells, even when the total soil moisture content appears adequate?

Options :

1. Soil temperature
2. Soil matric potential
3. Soil pH
4. Soil redox potential

Question Number : 51 Question Id : 4410092126153 Question Type : MCQ

Which of the following best explains the primary ecological role of soil microorganisms in nutrient cycling within terrestrial ecosystems?

Options :

1. They directly fix atmospheric carbon into soil organic matter for plant use.
2. They primarily store water and regulate soil moisture for ecosystem balance.
3. They absorb solar energy and convert it to chemical energy for food webs.
4. They decompose organic matter and release essential nutrients for plant uptake.

Question Number : 52 Question Id : 4410092126161 Question Type : MCQ

Which process carried out by soil microorganisms plays a direct role in returning carbon dioxide to the atmosphere during the carbon cycle?

Options :

1. Methanogenesis in wetlands
2. Nitrogen fixation in root nodules
3. Decomposition of organic matter
4. Photosynthesis by cyanobacteria

Question Number : 53 Question Id : 4410092126148 Question Type : MCQ

Two soil microbes coexist in anaerobic conditions; neither grows alone, but together they thrive because one removes a toxic metabolic by-product for the other. This obligate interaction is called:

Options :

1. Commensalism
2. Synergism
3. Syntrophism
4. Amensalism

Question Number : 54 Question Id : 4410092126150 Question Type : MCQ

Which soil fraction most effectively protects extracellular microbial enzymes from proteolytic degradation?

Options :

1. Sand
2. Silt
3. Clay minerals
4. Soil air

Question Number : 55 Question Id : 4410092126152 Question Type : MCQ

A sulfate-reducing bacterium consumes hydrogen produced by a fermentative bacterium, making fermentation energetically favorable. This interaction is best described as:

Options :

1. Commensalism
2. Syntrophic mutualism
3. Amensalism
4. Competition

Question Number : 56 Question Id : 4410092126158 Question Type : MCQ

Which evolutionary mechanism explains the loss of fitness in ancestral environments following a bacterium's strict niche specialization?

Options :

1. Antagonistic pleiotropy
2. Homologous recombination
3. Spontaneous mutagenesis
4. Horizontal gene transfer

Question Number : 57 Question Id : 4410092126160 Question Type : MCQ

Which genetic element enables bacteria to capture and express multiple resistance gene cassettes via site-specific recombination, making it a key driver of adaptive genome plasticity in environmental bacteria?

Options :

1. Insertion sequence (IS) elements
2. Conjugative transposons
3. Class 1 integrons
4. Pathogenicity islands

Question Number : 58 Question Id : 4410092126168 Question Type : MCQ

A PGPR promotes plant growth under drought stress by cleaving 1-aminocyclopropane-1-carboxylate (ACC) to lower ethylene levels in roots. Which specific enzyme is responsible for this cleavage?

Options :

1. ACC oxidase
2. ACC deaminase
3. ACC synthase
4. ACC synthase kinase

Question Number : 59 Question Id : 4410092126370 Question Type : MCQ

When sugarcane is used as a raw material for anaerobic fermentation by yeast during biofuel production, which of the following is the main product formed?

Options :

1. Penicillin
2. Ethanol
3. Chlorophyll
4. Calcium carbonate

Question Number : 60 Question Id : 4410092126381 Question Type : MCQ

Choose the **CORRECT** option.

Assertion (A): Foodborne pathogens can cause disease even when food appears normal.

Reason (R): Pathogens do not always produce detectable spoilage changes in food.

Options :

1. Both A and R are true, and R is the correct explanation of A.
2. Both A and R are true, but R is not the correct explanation of A.
3. A is true, but R is false.
4. A is false, but R is true.

Question Number : 61 Question Id : 4410092126384 Question Type : MCQ

Why is serial dilution commonly performed before plating a food sample?

Options :

1. To increase microbial growth rate
2. To obtain countable colonies
3. To sterilize the sample
4. To identify bacterial species

Question Number : 62 Question Id : 4410092126372 Question Type : MCQ

A fermentation process converts ethanol into vinegar. Which microbial activity is essential for this conversion?

Options :

1. Reduction of pyruvate to ethanol
2. Oxidation of ethanol to acetic acid
3. Hydrolysis of starch to glucose
4. Conversion of methane to carbon dioxide

Question Number : 63 Question Id : 4410092126383 Question Type : MCQ

Consumption of food contaminated with enteric pathogens commonly results in _____.

Options :

1. Improved digestion
2. Increased appetite
3. Gastrointestinal symptoms
4. Enhanced immunity

Question Number : 64 Question Id : 4410092126386 Question Type : MCQ

All of the following methods are used to quantify microorganisms in food EXCEPT _____.

Options :

1. Standard plate count
2. Most Probable Number
3. Membrane filtration
4. Gram staining

Question Number : 65 Question Id : 4410092126389 Question Type : MCQ

HACCP, a preventive food safety management system, stands for _____.

Options :

1. Hazard Assessment and Critical Control Procedure
2. Hazard Analysis and Critical Control Point
3. Health Analysis and Consumer Control Program
4. Health Assessment and Critical Check Point

Question Number : 66 Question Id : 4410092126371 Question Type : MCQ

Which of the following is NOT an advantage of biofuels?

Options :

1. Renewable source of energy
2. Reduced dependence on fossil fuels
3. Generation from biological materials
4. Unlimited energy output without biomass input

Question Number : 67 Question Id : 4410092126374 Question Type : MCQ

Which of the following statements CORRECTLY explains why high-moisture foods spoil rapidly?

Options :

1. Water supports microbial metabolism and growth.
2. Water prevents microbial enzyme activity.
3. Water destroys microbial cells.
4. Water blocks nutrient utilization.

Question Number : 68 Question Id : 4410092126380 Question Type : MCQ

Which of the following factors has the greatest influence on microbial destruction during heat treatment?

Options :

1. Temperature and exposure time
2. Food color and flavor
3. Packaging design only
4. Atmospheric pressure only

Question Number : 69 Question Id : 4410092126382 Question Type : MCQ

Which of the following is the most effective way to interrupt the transmission of many foodborne pathogens?

Options :

1. Maintaining proper hygiene during food handling
2. Increasing food moisture content
3. Storing food at room temperature
4. Delaying food processing

Question Number : 70 Question Id : 4410092126390 Question Type : MCQ

Which primary regulatory body monitors GMP compliance and food safety standards in the United States?

Options :

1. WHO
2. FDA
3. ISO
4. EMA

Question Number : 71 Question Id : 4410092126394 Question Type : MCQ

At which stage of drug development is a candidate drug first tested in human volunteers?

Options :

1. Drug discovery
2. Preclinical research
3. Phase I clinical trial
4. Phase III clinical trial

Question Number : 72 Question Id : 4410092126396 Question Type : MCQ

A researcher compares a new antihypertensive drug with an inactive tablet that resembles the drug in appearance. The inactive tablet is known as a/an _____.

Options :

1. Vaccine
2. Adjuvant
3. Placebo
4. Biomarker

Question Number : 73 Question Id : 4410092126401 Question Type : MCQ

What is the term for the fully assembled and infectious form of a virus, consisting of its nucleic acid enclosed within a capsid?

Options :

1. Capsid
2. Capsomere
3. Virion
4. Viroid

Question Number : 74 Question Id : 4410092126406 Question Type : MCQ

The genome of most DNA animal viruses is replicated primarily in which compartment of the host cell?

Options :

1. Lysosome
2. Mitochondria
3. Nucleus
4. Ribosomes

Question Number : 75 Question Id : 4410092126404 Question Type : MCQ

The Baltimore classification scheme specifically groups viruses according to _____.

Options :

1. The shape of their capsid only
2. How their mRNA is produced
3. The specific species of host they infect
4. The presence or absence of an envelope

Question Number : 76 Question Id : 4410092126411 Question Type : MCQ

What is the defining feature of a positive-sense (+) viral RNA strand?

Options :

1. It undergoes reverse transcription before expression
2. It functions directly as messenger RNA
3. It serves only as a complementary RNA template
4. It is packaged only as multiple RNA segments

Question Number : 77 Question Id : 4410092126414 Question Type : MCQ

The fusion of adjacent host cells induced by the measles virus results in the formation of a _____.

Options :

1. Prophage
2. Capsomere
3. Syncytium
4. Transformant

Question Number : 78 Question Id : 4410092126405 Question Type : MCQ

Which of the following is an example of a viral genus name?

Options :

1. Herpesviridae
2. Lentivirus
3. Picornaviridae
4. Herpesales

Question Number : 79 Question Id : 4410092126409 Question Type : MCQ

Herpesviruses initiate infection of host cells through _____.

Options :

1. Injection of viral DNA into the host cell
2. Fusion of the viral envelope with host cell membrane
3. Diffusion of viral particles through membrane
4. Integration of viral DNA into host genome

Question Number : 80 Question Id : 4410092126412 Question Type : MCQ

Before viral proteins can be synthesized in a negative-sense (–) RNA virus, what must occur first?

Options :

1. The viral RNA must integrate into host DNA.
2. The viral RNA must attach to ribosomes.
3. The viral RNA must convert into dsDNA.
4. The viral RNA must be copied into +RNA.

Question Number : 81 Question Id : 4410092126425 Question Type : MCQ

Besides mosquito bites, which route can also transmit the Zika virus?

Options :

1. Unpasteurized milk
2. Aerosol inhalation
3. Sexual contact
4. Soil contact

Question Number : 82 Question Id : 4410092126427 Question Type : MCQ

Which virus is most strongly associated with fruit bats as its natural reservoir?

Options :

1. Poliovirus
2. Ebolavirus
3. Rhinovirus
4. Adenovirus

Question Number : 83 Question Id : 4410092126665 Question Type : MCQ

Which enzyme is essential for joining DNA fragments during cloning?

Options :

1. Type II restriction endonucleases
2. DNA polymerase
3. Alkaline phosphatase
4. DNA ligase

Question Number : 84 Question Id : 4410092126677 Question Type : MCQ

Which process uses microbes to extract metals from ores?

Options :

1. Biomining
2. Fermentation
3. Composting
4. Bioremediation

Question Number : 85 Question Id : 4410092126667 Question Type : MCQ

Which process is most essential for introducing targeted gene edits into a bacterial genome using recombinant DNA technology?

Options :

1. Translation of Histone genes
2. Slicing of housekeeping genes
3. Homologous recombination
4. Random mutagenesis

Question Number : 86 Question Id : 4410092126679 Question Type : MCQ

Which combination of microorganisms is most effective for phosphate solubilization in acidic soils?

Options :

1. Azospirillum and Nostoc
2. Rhizobium and Frankia
3. Azotobacter and Bacillus
4. Pseudomonas and Bacillus

Question Number : 87 Question Id : 4410092126685 Question Type : MCQ

Which feature distinguishes bioplastics from microbial origin over petroleum-based plastics in waste management?

Options :

1. Biodegradable nature
2. Packaging use
3. Mechanical strength
4. Recyclability

Question Number : 88 Question Id : 4410092126666 Question Type : MCQ

A researcher clones a eukaryotic gene directly from genomic DNA into a bacterial expression vector but detects no functional protein despite successful transformation. Which factor most likely explains this outcome?

Options :

1. Presence of high-copy plasmids preventing accurate replication of recombinant DNA
2. Presence of bacterial transcription terminators within the coding sequence only
3. Presence of introns that cannot be correctly processed by bacterial cells
4. Presence of multiple ribosome-binding sites causing excessive translation initiation

Question Number : 89 Question Id : 4410092126678 Question Type : MCQ

In microbial biomining, controlling the pH of the leaching environment is essential for optimal metal recovery because it _____.

Options :

1. Increases the growth of microorganisms.
2. Enhances the solubility of target metal.
3. Neutralizes the byproducts.
4. Accelerates all the reactions.

Question Number : 90 Question Id : 4410092126680 Question Type : MCQ

A field treated with a Rhizobium biofertilizer fails to show expected results. Which is the most plausible explanation?

Options :

1. The field received adequate irrigation throughout the growing season
2. The crop was exposed to sufficient sunlight during its growth period
3. The soil contained adequate organic matter.
4. The crop is a non-legume crop unfit for symbiosis

Question Number : 91 Question Id : 4410092126686 Question Type : MCQ

A waste management company seeks a bioplastic that decomposes without industrial composting. Which **microbial biopolymer** is most suitable?

Options :

1. PCL (Polycaprolactone)
2. PET (Polyethylene terephthalate)
3. PHB (Polyhydroxybutyrate)
4. PLA (Polylactic acid)

Question Number : 92 Question Id : 4410092126687 Question Type : MCQ

Which design principle in synthetic biology ensures modular assembly of biological parts for predictable behavior in engineered organisms?

Options :

1. Diversification
2. Standardization
3. Optimization
4. Integration

Question Number : 93 Question Id : 4410092126699 Question Type : MCQ

Which of the following analytical methods is suitable for identifying a wide range of small molecules in microbial samples?

Options :

1. DNA hybridization
2. Mass spectrometry
3. Staining with different stains
4. Southern blotting

Question Number : 94 Question Id : 4410092126692 Question Type : MCQ

In phylogenetic analysis of microbes, which alignment type is crucial to accurately compare homologous genetic sequences across species?

Options :

1. Gapless Sequence Alignment
2. Multiple Sequence Alignment
3. Pairwise Sequence Alignment
4. Random Sequence Alignment

Question Number : 95 Question Id : 4410092126698 Question Type : MCQ

A researcher wants to compare protein expression profiles between two bacterial strains under stress. Which method provides the most comprehensive analysis?

Options :

1. Whole Genome Sequencing
2. Shotgun proteomics
3. Edman degradation
4. RNA Sequencing (RNA-seq)

Question Number : 96 Question Id : 4410092126701 Question Type : MCQ

A researcher identifies novel genes in soil metagenomes but struggles to assign function. Which advanced method could best resolve this issue?

Options :

1. Sequence alignment
2. Phylogenetic tree
3. Gene context
4. Motif search

Question Number : 97 Question Id : 4410092126710 Question Type : MCQ

Which principle is essential in bioethics for microbial research involving human subjects?

Options :

1. Profit sharing
2. Informed consent
3. Non-confidentiality
4. Material transfer

Question Number : 98 Question Id : 4410092126705 Question Type : MCQ

Which molecular diagnostic technique is preferred for detecting low-abundance pathogens rapidly in clinical samples due to its high sensitivity and specificity?

Options :

1. Conventional PCR
2. Sanger sequencing
3. DNA microarray
4. Real-time PCR

Question Number : 99 Question Id : 4410092126708 Question Type : MCQ

Which gene editing method allows for precise base changes without introducing double-strand breaks, when editing a gene responsible for disease resistance?

Options :

1. TALENs
2. Base editors
3. Zinc fingers
4. CRISPR-Cas9

Question Number : 100 Question Id : 4410092126711 Question Type : MCQ

A multinational microbial research project involves laboratories from several countries with different biosafety regulations. Which approach would best ensure consistent and responsible biosafety practices across all participating institutions?

Options :

1. Voluntary codes
2. Harmonized standards
3. Local adaptation
4. Bilateral agreements